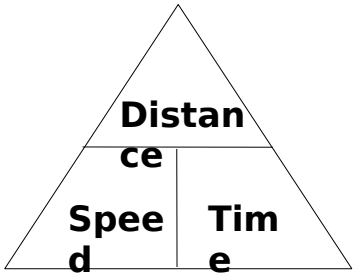
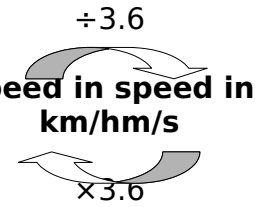
Speed and Velocity Practice Problems



Speed

1. If a car travels 500m in 40 seconds how fast is it going in m/s?
2. A cyclist travels 60km in 2 hours. What is his speed in km/h?
3. Tim drives 400m to the shops in 20 seconds.
   a. Calculate the average speed of the car in m/s.
   b. If the speed limit in the area is 70km/h, was Tim speeding?
4. The pitcher's mound in baseball is 85 m from the plate. It takes 4 seconds for a pitch to reach the plate. How fast is the pitch?
5. A train travels 510m in a time of 17 seconds.
   a. Calculate the average speed of the train in m/s.
   b. Calculate the average speed of the train in km/h.
6. A plane travels 395,000 meters in 9000 seconds.
   a. What was its speed in m/s?
   b. How fast is it travelling in km/h?



7. Barry walks to Scott's house, a distance of 600m in 5 minutes, but Scott is not home. Barry then walks to John's house, 300m away and it takes him 3 minutes.
   a. What speed, in m/s, was Barry walking at on his way to Scott's house?
   b. What was Barry's speed, in m/s, between Scott and John's house?
   c. What was Barry's average speed, in m/s, for the whole journey?

Time

1. How long does it take for a car travelling at 15m/s to go 600m?
2. How long will it take for a bug to travel 9 meters across the floor if it is travelling at 1.5 m/s?
3. A bus is travelling at 20m/s. How long does it take for the bus to travel 72km? (Hint: change the speed to km/h)
4. A bullet travels at 850 m/s. How long will it take the bullet to go 1 km? (Hint: change the distance into m)
5. Sarah swims backstroke at an average speed of 8 m/s. How long will it take her to complete a 200 m race?
6. How long would it take for the sound of thunder to travel 1, 500 m if the speed of sound is 340 m/s?
7. A train travels in a straight line from one station to the next at 40km/h and then returns at 60km/h. If the two stations are 5km apart, how long is the train's journey?

Distance

1. A model car travels at 2.5m/s for 5 seconds. How far does the car travel?
2. A train is moving at 10m/s. How far does it travel in 5 seconds?

3. The fastest train in the world moves at 500 km/h. How far will it go in 3 hours?
4. Mike rides his motorcycle at an average speed of 20 m/s for 500 seconds.
   a. How far did he ride in metres?
   b. How many kilometres did he ride?
5. A snail can move approximately 0.3 m/min. How many metres can the snail cover in 15 minutes?
6. A person in a kayak paddles down a river at an average speed of 10 km/h. After 3 hours, how far has she travelled?
7. A plane flies east at 250km/h for 3 hours and then south at 200km/h for 1 hour.
   a. How far did the plane travel in the first part of the flight?
   b. How far did the plane travel in the second part of the flight?
   c. What is the total distance travelled?
   d. What is the average speed of the plane for the whole journey?

Distance

Speed Time

Velocity (Optional)

1. A plane flies 420km due north in 2hours. Find the average velocity of the plane in km/h.
2. A boat sails 1.2km due east for a time of 5 minutes. Find the average velocity of the boat in m/s.
3. An aeroplane flies due north with an average velocity of 85m/s for 20s. Calculate the displacement of the aeroplane.
4. A truck travels along a straight road with a constant velocity of 30m/s. After 15 seconds, how far has the truck travelled?
5. Beth rides her horse 3km east and then 4km north. The trip takes her 30 minutes.
   a. What is the total distance Beth travelled?
   b. What is her average speed in km/h?
   c. What is Beth's displacement?
   d. What is her average velocity?
6. Amanda drives her car 35km north and then 75km south in a time of 2 hours.
   a. What is her average speed in km/h?
   b. What is her displacement?
   c. What is her average velocity in km/h?

CHALLENGE PROBLEM 1

Bill and Amy want to ride their bikes from their house to school which is 14.4 kilometers away. It takes Amy 40 minutes to arrive at school. Bill arrives 20 minutes after Amy. How much faster (in meters/second) is Amy's average speed for the entire trip?

CHALLENGE PROBLEM 2



Do you remember the fable of the tortoise and the hare? The giant tortoise can reach a speed of about 7.5cm/s. A hare can run as fast as 20m/s. If their race was 1km long, how long was the hare asleep if the race resulted in a tie?

÷3.6

speed in speed in km/hm/s

×3.6